

# Yutao Xie

 YutaoXie |  yux076@ucsd.edu |

## Education

### University of California, Santa Barbara

Ph.D. in Computer Science  
Advised by Prof. Shiyu Chang

*Santa Barbara, CA, USA*

*Incoming Sep. 2026*

### University of California, San Diego

M.S. in Computer Science, GPA: 3.925/4.0  
Advised by Prof. Xiaolong Wang and Prof. Zhiting Hu

*San Diego, CA, USA*

*Sep. 2024 – Present*

### Tianjin University

B.E. in Software Engineering, GPA: 3.85/4.0

*Tianjin, China*

*Aug. 2020 – Jul. 2024*

## Publications

### IsoCompute Playbook: Optimally Scaling Sampling Compute for RL Training of LLMs

International Conference on Machine Learning (ICML)

Zhoujun Cheng\*, **Yutao Xie\***, Yuxiao Qu\*, Amrith Setlur\*, Shibo Hao, Varad Pimpalkhute, Tongtong Liang, Feng Yao, Zhengzhong Liu, Eric Xing, Virginia Smith, Ruslan Salakhutdinov, Zhiting Hu, Taylor Killian, Aviral Kumar

[Blog](#) | [Arxiv](#)

*TL; DR:* A compute-optimal workflow for scaling on-policy RL sampling in LLM post-training.

*Under review*

*2026*

### TIPS: Turn-Level Information–Potential Reward Shaping for Search-Augmented LLMs

International Conference on Learning Representations (ICLR)

**Yutao Xie**, Nathaniel Thomas, Nicklas Hansen, Yang Fu, Erran Li, Xiaolong Wang

[Openreview](#)

*TL;DR:* A self-distillation method leveraging mutual-information-based reward for finer credit assignment of RLVR.

*Poster*

*2026*

### K2-Think: A Parameter-Efficient Reasoning System

ArXiv preprint

Institute of Foundation Models, MBZUAI, [...], **Yutao Xie**, [...] (31 authors)

[Arxiv](#) | [Project](#) / [Models](#)

*TL;DR:* K2-Think is a 32B model that rivals much larger ones via six training/inference technique pillars.

*Tech Report / Model Release*

*2025*

### Revisiting Reinforcement Learning for LLM Reasoning from a Cross-Domain Perspective

Conference on Neural Information Processing Systems (NeurIPS)

Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, **Yutao Xie**, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, Zhiting Hu

[Arxiv](#)

*TL;DR:* A cross-domain RL dataset with analysis showing domain-dependent RL gains across six reasoning domains.

*Poster*

*2025*

### Sound Identification of Abnormal Pig Vocalizations: Enhancing Livestock Welfare Monitoring on Smart Farms

Information Processing & Management (IPM)

**Yutao Xie**, Jun Wang, Cheng Chen, Taixin Yin, Shiyu Yang, Zhiyuan Li, Ye Zhang, Juyang Ke, Le Song, Lin Gan

[Journal](#)

*TL;DR:* A DDP inference system loaded with an AST-based detector method to help monitoring in smart farms.

*Journal*

*2024*

## Research Experience

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### MixLab

Research Assistant, advised by Prof. Zhiting Hu

San Diego, CA, USA

May. 2025 – Present

- Helped MBZ-IFM to build **K2-think model**. Curated Guru data; evaluated and analyzed performance of checkpoints on Guru/math (incl. Pass@K). Built a semantic de-duplication pipeline for the incoming dataset.
- Studied **compute-optimal RL sampling** for LLM post-training; characterized scaling trends across parallel rollouts ( $n$ ), problems per batch ( $B_{problem}$ ), and sequential iterations ( $M$ ), analyzed gradient and critical batch size in RLVR, in collaboration with MBZUAI-IFM and Prof. Aviral Kumar (CMU-AIRe Lab).

### WangLab

Research Assistant, advised by Prof. Xiaolong Wang and Prof. Li Erran Li

San Diego, CA, USA

Nov. 2024 – Present

- For LLM with **Search Engines** on QA tasks, proposed and implemented **TIPS** (Turn-level Information–Potential Reward Shaping) to improve training effectiveness and stability.
- Built tooling for **Text-to-SQL** and **CodeGen**—a direct-DB interface and a Jupyter-like executor—and trained the LLM to use them with **RLVR**.
- Developed a **model-based RL** method for discretized control on top of TDMPC2, incorporating planners (MCTS, CEM) with **world models** on DMControl and Atari; used in **Musculoskeletal Control with TD-MPC2**

## Work Experience

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### AutoBrain Science and Technology Co., Ltd

Machine Learning Engineer (Intern)

Tianjin, China

Apr 2023 – Jul 2023

- Trained and deployed object detection CV models on Jetson AGX Xavier; created a perf benchmark (throughput, latency, memory) and optimized model inference by quantization and pruning.

## Awards and Scholarships

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### 2024 Outstanding Graduation Thesis (Top 1%)

- Optimizing Supply Chain Contract Negotiation Based on Multi-agent Reinforcement Learning.

Award

### 2022 Bronze Prize — China International “Internet+”

- Project: Calino — AI Monitoring and Data Mining System.

Award

### 2022/23 “Three-Good (Sanhao) Student” Scholarship

- A Merit scholarship for moral character, academics, and physical fitness.

Scholarship

### 2021 Outstanding Student Leadership Scholarship

- A scholarship recognizing excellence in student leadership, service, and impact.

Scholarship

## Academic Service

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### Conference and workshop reviewer

Conference on Neural Information Processing Systems (**NeurIPS**)

2025

International Conference on Learning Representations Workshop on Scaling Post-training for LLMs (**ICLR**)

2026

### Journal reviewer

Applied Energy (**AE**)

2025

Information Processing & Management (**IP&M**)

2024